$H(X) = \Sigma -p_i \log p_i = E(-\log p)$ $h(X) = \int -p \log p$ $h(X)$ can < 0

$I(X;Y) = \iint p_{XY} \log \frac{p_{XY}}{p_X p_Y} = E(\log \frac{p_{XY}}{p_X p_Y})$ $I \geq 0$ Pf: $-I(X;Y) = \sum_x \sum_y p(x,y) \log \frac{p(x)p(y)}{p(x,y)} \leq \sum_x \sum_y p_{XY} \frac{1}{\ln^2}[\frac{p(x)p(y)}{p(x,y)} - 1] = 0$

Data proc ineq $I(X;Y) \geq I(X;Z)$ Chain Rule $H(X,Y) = H(X) + H(Y|X)$ $H(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \cdots) = \sum_{n=1}^{\infty} \frac{n}{2^n} = 2$

$H(p)$ concave func $H'(p) = \log \frac{1-p}{p}$ $H''(p) = \frac{1}{\ln^2 p(p-1)} < 0$ $\sum_{n=0}^{\infty} x^n = \frac{1}{1-x}$ $\sum_{n=1}^{\infty} x^n = \frac{x}{1-x}$ $\sum_{n=1}^{\infty} n x^{n-1} = \frac{1}{(1-x)^2}$

$g: X \to Y$ $H(g(X)) \leq H(X)$ $g: X \leftrightarrow Y$ $H(g(X)) = H(X)$ e.g. $H(R) \leq H(R, X_n) = H(X^n) \leq H(R) + 1$

X, Y indp $\Rightarrow H(X+Y) \geq H(X)$ Pf: $H(X) = H(X|Y) = H(X+Y|Y)$ $E(\log X) \leq \log E(X)$

$H(X+Y) = H(X) + H(Y) \Leftrightarrow X, Y$ indp and $(X,Y) \leftrightarrow X+Y$ Pf: $H(X+Y) \leq H(X,Y) \leq H(X) + H(Y)$

e.g. Bottleneck $X \to Y \to Z$, $I(X;Z) \leq \log|Y|$ Pf: $I(X;Z) \leq I(X;Y) = H(Y) - H(Y|X) \leq H(Y)$

e.g. $H(X|g(Y)) = H(X|Y) \Leftrightarrow X \to g(Y) \to Y$ …

Entropy Rate $H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X^n) = \lim_{n\to\infty} H(X_n|X^{n-1})$ $H(\mathcal{X}) \leq H(X_0)$ Pf: $H(\mathcal{X}) = H(X_0|X_{-1}$

e.g. $\frac{1}{2n} I(X_1 \cdots X_n; X_{n+1} \cdots X_{2n}) = \frac{1}{2n}[2H(X^n) - H(X^{2n})] \to H(\mathcal{X}) - H(\mathcal{X}) = 0$ st Markov $H(\mathcal{X}) = H(X_2|X_1) = \sum \pi^j H(j)$

异值域 $X \begin{cases} \sim P_1(1,2,3), & \alpha \\ \sim P_2(4,5), & 1-\alpha \end{cases}$ $H(X) = H(X,C) = H(C) + H(X|C) = H(\alpha) + \alpha H(P_1) + (1-\alpha) H(P_2)$

同值域 $Y_i \begin{cases} X_i \sim Ber(P_1), & \frac{1}{2} \\ X_i \sim Ber(P_2), & \frac{1}{2} \end{cases}$ 若C每次随机 $Y_1 \cdots Y_N$ indp $H(Y) = H(Y_1) = H(\frac{1}{2}(P_1 + P_2))$

若C选一次就不变了 $Y_1 \cdots Y_N$ not indp $H(Y) = \lim_{n\to\infty} \frac{1}{n}[\cancel{H(C)} + H(Y|C) - \cancel{H(C|Y)}]$

$= \frac{1}{2}(H_1 + H_2)$

Prefix / Instantaneous $\subset$ Uniquely decodable $\subset$ Non-singular

0, 10, 110, 111 $X^n \leftrightarrow Z^n$ 10, 00, 11, 110 $X \leftrightarrow Z$ 0, 00, 000

$H(C(X^n)) = H(X^n)$, $H(\mathcal{C}(X)) = H(\mathcal{X}) / E(L)$

X	1	2	3
P	1/2	1/4	1/4
C	00	1	01

$H(\mathcal{X}) = H(X) = \frac{3}{2}$

$\bar{L} = 2 \times \frac{1}{2} + 1 \times \frac{1}{4} + 2 \times \frac{1}{4}$

$H(\mathcal{C}(X)) = H(X) / \bar{L}$

Opt coding min $L = E(l(x)) = \sum p(x) l(x)$

s.t. $l(x) \in \mathbb{Z}$, $\sum 2^{-l(x)} \leq 1$ Kraft ineq

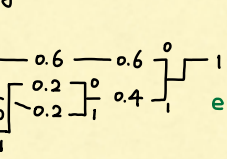
prefix $\Rightarrow$ Kraft $\Rightarrow$ we can construct prefix $\min \bar{L}_{prf} = \min \bar{L}_{uni\text{-}dcd}$

$H(X) \leq L^* < H(X) + 1$

$L_n^* = \frac{1}{n} E(l(x^n)) \to H(\mathcal{X})$ iid $H(X) \leq L_n^* < H(X) + \frac{1}{n}$

Huffman Coding

C	X	P
0	1	0.6
11	2	0.2
100	3	0.1
101	4	0.1



Prop. $H(X) \leq L^H = L^* < H(X) + 1$ $L^H(x)$ 不一定

$P(x) \uparrow$ $l(x) \downarrow$ P_{max} 和 $P_{2nd\text{-}max}$ 差1位

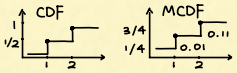
$C^* \not\Rightarrow C^H$ e.g. $(\frac{1}{3}, \frac{1}{3}, \frac{1}{4}, \frac{1}{12})$ Huffman $l = (1, 2, \underline{3}, 3)$ $> \lceil -\log \frac{1}{4} \rceil$ 另 $l = (2,2,2,2)$ 也最优

e.g. $X \sim U[1, \cdots, n]$, $L^H(X) = \lfloor \log n \rfloor$ or $\lceil \log n \rceil$ Depth $\lceil \log n \rceil$

deepest nodes $2(n - 2^{\lfloor \log n \rfloor})$

Shannon Coding $l^S(x) = \lceil -\log P(x) \rceil$ not opt $L^S < H(X) + 1$

SFC $l^{SFC}(x) = \lceil -\log P(x) \rceil + 1$ $L^{SFC} < H(X) + 2$



X	0	1
P	1/2	1/2
C^H	0	1
C^{SFC}	01	11

$C^H(X)$ binary tree

Lempel-Ziv Coding: Given aaabaabaabaab …… Decoding: Given 132461

initial a b → a b aa aab ba aaba aabaa

1 2 3 4 5 6 7

加上下一位是aa，表中加入aa为3

aaa 从表中读取得 aa=3

initial a b aaa b aab aaba aa

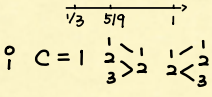
1 2 3 4 5 6 7

132… a ? 不知道3表示什么，初始化中没有3

但一定是前位加出来的，故 a?

a a? 故3为aa a a a

Arithmetic Coding $X_i \sim B(\frac{2}{3})$, $F(101) = \frac{13}{27}$



$F(01101) = F(00) + F(010) + F(01100) = (\frac{1}{3})^2 + \frac{2}{3} \cdot (\frac{1}{3})^2 + (\frac{2}{3})^2 \cdot (\frac{1}{3})^3$

Capacity $C = \max I(X;Y)$ $1 \to 1$ $C = 0$ $C = 1$ $C = \log 2$ $C = \log 3 - 1$

e.g. $Z \sim U(M)$ $C = \log \frac{N}{M}$

$X \in \{1 \cdots N\} \to \oplus \to Y = X + Z \pmod N$

BSC $C = 1 - H(P)$ $Z \sim Ber(P)$, $Y = X + Z$

BEC $C = 1 - P$ Pf: $I(X;Y) = H(Y) - H(Y|X) \leftarrow H(P)$

$H(Y,E) = H(E) + H(Y|E) = H(P) + (1-P)H(X)$

e.g. $Z \sim U(3)$ $P(X) = (\frac{1}{2}, 0, \frac{1}{2})$

$X \in \{1,2,3\} \to \oplus \to Y$ $P(Y) = (\frac{1}{6}, \frac{1}{6}, \frac{1}{3}, \frac{1}{6}, \frac{1}{6})$ $C = \frac{2}{3}$

Z shape $C = \log(2^{H/P} + 1) - H(P)/P$ Pf: $I(X;Y) = H(Y) - H(Y|X)$

$= H(\pi p) - \pi H(p)$, $d/d\pi$

$p \log \frac{1-\pi p}{\pi p} - H(p) = 0$,

$\pi = [p(2^{H/P} + 1)]^{-1}$

Weakly Symmetric $C = \log 2 - H(\frac{1}{2}, \frac{1}{3}, \frac{1}{6})$

e.g. $Z \sim U\{0, a\}$ $C = \begin{cases} 1/2, & a = \pm 1 \\ 1, & \text{else} \end{cases}$

$X \in \{0, 1\} \to \oplus \to Y$

e.g. 时序并联 $X_1 - C_1 - Y_1$

$(Z_1, Z_2$ 可相关$)$ $X_2 - C_2 - Y_2$

e.g. 多观测 $X - C \begin{cases} Y_1 \\ Y_2 \end{cases}$

$(X, Z$ 可相关$)$ Assume $I(Y_1, Y_2 | X) = 0$

$C \geq C_1 + C_2$ When Z_1, Z_2 indp $C = C_1 + C_2$

Pf: $I(X_1, X_2; Y_1, Y_2) = H(X_1, X_2) - H(X_1, X_2 | Y_1, Y_2) \leq H(Z_1, Z_2)$

When X indp, $\max H(X_1, X_2) = H(X_1) + H(X_2)$ So $C \geq C_1 + C_2$

If Z_1, Z_2 indp $I(X_1, X_2; Y_1, Y_2) = H(Y_1, Y_2) - H(Y_1, Y_2 | X_1, X_2)$

$\leq H(Y_1) + H(Y_2)$, so $C = \max_{P_1, P_2} I = C_1 + C_2$

$C \leq 2C_1$ When Y_1, Y_2 indp $C = 2C_1$

Pf: $I(X; Y_1, Y_2) = I(X; Y_1) + I(X; Y_2) - I(Y_1; Y_2)$

$C = \max I(X; Y_1, Y_2) \leq \max I(X; Y_1) + \max I(X; Y_2)$

$= 2H(Y_1 | X_1)$

e.g. 串联 $\forall C_1 + \text{BEC}$, $C = C_1(1-\alpha)$

Pf: $I(X;Z) = H(Z) - H(Z|X) = H(Z,E) - H(Z,E|X) = (1-\alpha) I(X;Y)$

$H(E) + H(Z|E)$ $\quad H(\alpha) + (1-\alpha) H(Y|X)$

$H(\alpha) + (1-\alpha) H(Y)$

e.g. 选择型并联 $x \begin{cases} x_1 \to \boxed{C_1} \to y_1 \\ x_2 \to \boxed{C_2} \to y_2 \end{cases}$ $\quad 2^C = 2^{C_1} + 2^{C_2}$

$C = \log(2^{1-H} + 1)$

Pf: Let $P(x_1) = \theta$, $I(X;Y) = I(X;\theta) + I(X;Y|\theta) = H(\theta) + \theta I(x_1;y_1) + (1-\theta) I(x_2;y_2)$

$C = \max_\theta H(\theta) + \theta C_1 + (1-\theta) C_2 \qquad \theta^* = 2^{C_1}/(2^{C_1} + 2^{C_2})$

Gaussian Channel $h(X) = \frac{1}{2}\log 2\pi e \sigma^2$ cnt. $X \to Y$ noiseless, $C = \infty$

Pf: $h(X) = -\int \phi \log \phi = -\int \phi [-\log\sqrt{2\pi}\sigma - \frac{x^2}{2\sigma^2}\log e] = \log\sqrt{2\pi}\sigma \int \phi(x)dx + \log e \int \phi(x) \frac{x^2}{2\sigma^2} dx$

Gauss max Pf: $h(f) - h(\phi) = \int \phi \log \phi - f \log f = \int f \log \frac{\phi}{f} \le \log \int f \cdot \frac{\phi}{f} = 0$

$C = \max I(X;Y) = \frac{1}{2}\log(1 + \frac{P}{N})$ s.t. $EX^2 \le P$ $\quad$ Pf: $H(Y) - H(Y|X) \le \frac{1}{2}\log 2\pi e(P+N) - H(Z)$

$Z \sim N(0,N)$

$X \to \oplus \to Y$

$C = \max I(X;Y) = \sum \frac{1}{2}\log(1 + \frac{P_i}{N_i})$ s.t. $\sum P_i \le P$ $\quad$ Waterfilling $P_i = (\nu - N_i)^+$

Bandlimited $2W$ samples per sec. $C = \frac{1}{2}\log(1 + \frac{P/2W}{\frac{1}{2}N_0})$ bit / sample $\quad C = W\log(1 + \frac{P}{WN_0})$ bit / sec

$W \to \boxed{E} \to X^n \to \boxed{C} \to Y^n \to \boxed{D} \to \hat{W}$ $\quad$ Coding Rate $R = \frac{1}{n}\log M$ $\quad$ Memoriless: $P(y_n | x^n, y^{n-1}) = P(y_n | x_n)$

M cases $\mathcal{X} = \{0,1\}$ (M,n) or $(2^{nR}, n)$ code

error prob $\lambda_i = P(W \ne \hat{W} | W = i)$ $\quad \lambda = \max \lambda_i$ $\quad P_e = \frac{1}{M}\sum_{i=1}^{M} \lambda_i$

Fano's ineq: $H(X|Y) \le H(P_e) + P_e \log(|\mathcal{X}| - 1)$ $\quad P_e \ge H(X|Y) - 1 / \log(|\mathcal{X}| - 1)$

Pf: $H(X|E, \hat{X}) = P_e H(X|E=1, \hat{X}) + (1-P_e) \cancel{H}(X|E=0, \hat{X}) \le P_e \log(|\mathcal{X}| - 1)$

$H(E|\hat{X}) + H(X|E, \hat{X}) = H(X, E|\hat{X}) = H(X|\hat{X}) = H(X|Y)$ $\quad H(E|\hat{X}) \le H(P_e)$

Thm. $R < C \Rightarrow \exists (M,n)$ code s.t. $\max \lambda_i \to 0$ (achievable) $\Rightarrow R \le C$

for small $P_e = P(W \ne \hat{W})$, let $W \sim U[1, \cdots, 2^{nR}]$, $H(W) = nR$

$H(W) = H(W|Y^n) + I(W;Y^n)$ $\quad nR \le 1 + nRP_e + nC$ $\quad$ when $n \to \infty$, $R \le C$ $\quad P_e \ge 1 - \frac{C}{R} - \frac{1}{nR}$

$\le 1 + nRP_e \le I(X^n;Y^n) \le nC$

$I(X^n;Y^n) \le nC$ for DMC Pf: $H(Y^n) - \sum H(Y_i|X_i) \le \sum H(Y_i) - H(Y_i|X_i)$

Feedback $C_{FB} = C$

BSC feedback $\underbrace{X_1 \to C \to Y_1}_{X_2}$

$I(X^n;Y^n) = 1 + (n-2)H(P)$

Pf: $I(X^n;Y^n) = H(Y^n) - H(Y^n|X^n)$

$H(Y^n) = H(Y_1) + \underbrace{H(Y_2|Y_1)}_{=H(Y_2|X_2)} + \cdots = 1 + (n-1)H(P)$

$H(Y^n|X^n) = H(Y_1|X^n) + \underbrace{H(Y_2|Y_1, X^n)}_{=0} + \cdots = H(P)$

Distorsion $D = E[(X - \hat{X})^2]$

Rate $R = \max I(X;\hat{X})$ s.t. $E[(X - \hat{X})^2] \le D$

$X \sim U[-A, A]$ $\quad D = \frac{1}{12}\Delta^2 = \frac{1}{12}(\frac{2A}{2^n})^2$ output n bits

$X \sim N(0, \sigma^2)$ $\quad R = \begin{cases} \frac{1}{2}\log\frac{\sigma^2}{D}, & D \le \sigma^2 \\ 0, & D > \sigma^2 \end{cases}$ $\quad D = \sigma^2 2^{-2R}$